

Don't be too calm:

How does the combination of visual and textual frames influence expressions of anger in short video comment sections?

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Abstract

Short-video platforms combine visual and textual information. However, little is known about how these two forms of information work together to shape emotional responses in comment sections. This study examines Douyin videos about the debate over sealing drug-use records in China. The dataset includes 449 videos and 419,126 top-level comments. To analyze the content of these videos, the study classifies both visual and textual frames into three types: intensifying, informational, and mitigating frames. For the visual analysis, frames were extracted at fixed time intervals, and similar frames were grouped together. A large language model was then used to describe representative frames and support the coding of visual frames. For the textual analysis, speech, subtitles, on-screen text, and video descriptions were collected and coded using the same framework. A BERT-based model was used to identify expressions of anger in the comments. The results show that visual and textual frames often move in the same direction and that different frame combinations are associated with different levels of anger. Videos with intensifying frames tend to receive more angry comments, while videos that use mitigating frames in both visuals and text show the lowest level of anger. Both visual and textual frames have significant main effects on anger expression. However, their interaction is not significant. The study shows why visual and textual frames should be examined together and provides a practical method for studying visual framing in dynamic short-video content.

Keywords: short video, multimodal framing, computational communication methods, social media, sentiment analysis

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comment sections?**

Short video platforms have become an important field for the dissemination of public issues. Short video platforms have become important spaces for the communication and discussion of public issues (O'Brien et al., 2025; Yu et al., 2023). Many issues that originally belonged to policy discussions or institutional explanations often no longer circulate in a smooth information form after entering the platform, but are quickly transformed into content with emotional tension. Anger deserves particular attention because it is closely connected to perceived injustice and responsibility attribution (Kühne et al., 2015; Montada & Schneider, 1989). It can also involve moral condemnation and stronger preferences for punishment (Kühne & Schemer, 2015). On social media, moral-emotional language is especially likely to spread through social networks (Brady et al., 2017). Positive social feedback can further encourage users to express moral outrage in later posts (Brady et al., 2021).

Because of this, a seemingly simple but critical question arises: why do different short videos centered around the same public issue inspire angry reactions of disparate intensity in the comment section? If the controversial nature of the issue itself is already relatively well established, do differences in comment section sentiment still depend primarily on the issue itself, or are they more deeply influenced by the way in which the video content is organized? This question relates not only to how we understand the formation mechanism of public sentiment in the short-video era, but also to how content producers in the platform can shape the direction of discussion and emotional tone through multimodal design.

The issue of “sealing drug abuse records” in China provides an appropriate case study for observing this issue. In June 2025, the revised Law of the People's Republic of China on Penalties for Administration of Public Security introduced a rule requiring records of public security violations to be sealed. The revised law took effect on January 1, 2026 (Standing Committee of the National People's Congress, 2025). The controversy quickly gained widespread

online attention. According to Zhiwei Data (2025), its online influence ranked above 98 percent of the events recorded in the platform's database. In the social context of China's long history of tough anti-drug policies and the public's high sensitivity to the drug problem, the issue easily and quickly spilled over from institutional explanations to emotional debates about fairness and moral boundaries. However, as the later analysis will show, short videos on the topic did not elicit the same intensity of angry reactions.

Framing theory provides an important starting point for understanding this issue. However, in contrast to the role of framing on cognitive behavior, the relationship between framing and episodic emotional expression still lacks direct testing. Second, research in recent years has begun to recognize the importance of visual frames, and has also pointed out that the relationship between visuals and text can change communication effects (Dan, 2017). Most of the existing results focus on news pictures or graphic materials, and there is still a lack of adequate answers as to how visual frames are recognized in a continuous and dynamic object such as a short video, how visuals and text are combined, and whether such combinations further affect the expression of anger in the comment section.

It is at this theoretical gap that this paper attempts to contribute. Benefiting from advances in computational methods in the field of visual communication and the gradual maturation of the image recognition capabilities of large language models (LLMs) (Peng et al., 2024), this paper proposes an analytical method for measuring visual frames in dynamic videos. The method was tested on 449 Douyin short videos about the Chinese controversy over sealing drug-use records. The study identified three dominant visual frames for comparison, encoded the textual frames using the same computational method, and used machine learning to recognize expressions of anger in the comment sections of the relevant short videos. Anger expression was then used as the dependent variable in the regression analysis.

The data results show that visual frames are prioritized in emotion setting, and different combinations of visual and textual frames can significantly correspond to different levels of anger expression, and their consistency can influence the anger expression in the comment section,

which means multimodal relationships can explain the differences in the emotion of the comment section better than a single modality. The impact on audience emotions depends not only on what is shown or what is said, but also on how presentation and narrative are combined.

Literature Review

Public Issue Communication and Emotional Interaction in Short Video Platforms

In the era of new media communication, short videos have profoundly reshaped the discussion pattern of public issues and the path of public participation. The short video platform represented by TikTok has gradually become an important field for opinion generation and communication interaction (Yu et al., 2023). Unlike traditional news texts or simple graphic reports, short video platforms present an interactive ecology that is fragmented, entertaining, and easily driven by algorithmic logic, which makes it easier for public issues to be embedded in the daily information flow and to be paid attention to and consumed (Liang & Ye, 2025; O'Brien et al., 2025).

Influenced by the platform economy and algorithmic logic, short-video communicators are more inclined than traditional information media to package public issues as interactive objects that reflect their subjective intentions and compete for attention. Emotionally charged content is more attention-grabbing than rational, neutral, and objective commentary that retains the facts (Eberl et al., 2020; van Dijck et al., 2018). It has been shown that, in the face of social media platforms' distribution logic centered on attention and interaction, communication agents tend to consciously enhance the emotionality of their content, e.g., by using mild linguistic features to influence users to subscribe (V. Y. Chen, 2025), by matching anxious and angry tones to attract replies (Lee & Britt, 2024), and by inducing clicks (Zhang & Fu, 2022).

Although emotions are usually not considered a reliable basis for judgment in the Platonic tradition, but are instead understood as psychological forces that need to be moderated and guided by reason, modern cognitive psychology suggests that emotions can strongly affect the perception and evaluation of issues and events (Kühne et al., 2011) and shape information processing (Kühne & Schemer, 2015). Especially in relation to controversial public issues, audiences often encounter

relevant content not by first forming systematic and stable policy judgments, but by first generating emotional reactions of varying intensity (Hasell et al., 2025). In this context, it is crucial to understand whether and how communication actors intend to influence audience emotions.

With the deepening of emotion research, scholars have begun to compare the role of different emotion categories on communication effectiveness. Scholars have found that anger is particularly expressed when policies can be perceived as going against the will of the community at large (Wayne, 2023). On the one hand, anger is a high-arousal, strongly attributed discrete emotion that is usually closely associated with a sense of injustice, accountability, moral condemnation, and a tendency to punish, and is therefore more likely to be aroused in topics involving risk, misconduct, connivance, or institutional controversy (Montada & Schneider, 1989). On the other hand, compared with negative emotions such as sadness or helplessness, anger is more visible and transmissible, and is more likely to be transformed into direct comments, retweets, emotional mobilization, and even collective actions (Valenzuela et al., 2017; van Zomeren et al., 2004).

In the case of “sealing drug-use records in China,” which involves both institutional boundaries and counterarguments, examining how the content of the record triggers anger is a key way to understand the consequences of its dissemination. At the same time, different short videos about the same public issue can stimulate anger reactions of different intensities. That is, explaining how different short videos about the same issue can stimulate different levels of anger among audiences in the same spatial and temporal environment has become an important research problem.

The key lies not only in whether the issue itself is controversial, but also in how the communicator constructs the issue emotionally through specific forms of content organization. The degree of anger triggered by short videos may therefore be closely related to how the content is integrated.

Framing theory states that the media do not mechanically transmit facts, but rather,

through the selection, highlighting and omission of specific information elements, they construct a specific meaning structure and guide the audience to understand and interpret it in its preconceived way, which promotes a specific definition of the problem, causal explanation, moral evaluation and treatment suggestions, thus influencing the audience's perceptions, and even their emotions and actions (Entman, 1993). In other words, if the emotional responses elicited by similar short videos show hierarchical and systematic differences, it is necessary for such differences to be traced back to the content structure of the video itself, i.e., what kind of frames it consists of.

Framing Effects and Anger Expression

Since Goffman proposed the concept of “framing,” framing theory has gradually become an important theoretical resource in communication studies for explaining how the media construct reality and shape meaning (Goffman, 1974). Framing theory helps explain the process by which media construct social reality and their limited role in shaping public perceptions and attitudes toward specific events or issues. In the case of controversial reporting, for example, the media can employ either relatively benign frames or frames that are more threatening and blame-oriented (Ekström & Shehata, 2024). In technology coverage, for example, the media may adopt a “progress frame” that highlights innovation and economic efficiency, a “risk frame” that emphasizes warnings and ethical concerns, or a “competition frame” that foregrounds international rivalry, thereby shaping differentiated public perceptions of technology (Weaver et al., 2009). Extending this to self-media platforms, framing remains a core concept for describing how creators influence public discourse.

In terms of media psychology, when audiences are exposed to a particular frame, that frame can activate existing cognitive schemas, heighten the psychological availability and salience of emphasized information, and thereby influence audiences' sensations, emotions, cognition, and behavior (Geise & Xu, 2025). Its emotional effects, especially on anger, operate mainly through three pathways: responsibility attribution, risk construction, and victim presentation. When content explicitly indicates who is responsible or who caused the harm,

audiences are more likely to form moral condemnations and, in turn, experience anger (Kühne et al., 2015). When an issue is presented as involving high threat and uncertainty, audiences are more likely to enter a state of high alert, which provides a basis for anger (Nabi, 2003). Visual depictions of victims, meanwhile, can intensify perceptions of injustice and harm, thereby strengthening anger toward the party deemed responsible (Dawtry et al., 2020). These three pathways intertwine within the same communicative unit and jointly shape the direction and intensity of audience emotions. However, as related reviews have noted, although empirical studies have begun to pay attention to the affective dimension of visual framing, much of the existing literature, including the studies mentioned above, has not treated audiences' affective responses as core outcome variables in their own right, but instead has regarded emotion as a mediating variable leading to subsequent effects such as cognition (Xu, 2025). In other words, there is still a lack of fuller empirical tests of how framing directly relates to audiences' outward emotional expression, especially in platform-based interactive environments.

In the age of social media, comment sections provide a relatively visible arena for observing such emotional responses. Although most viewers of video content tend to remain silent and commenters cannot fully represent the entire audience (Sun et al., 2014), comment sections are still among the few spaces on platforms where outward emotional expression becomes concentrated and observable. Therefore, this paper treats anger expressed in comment sections as an observable indicator of framing's emotional effects and uses it to examine whether different ways of organizing content systematically correspond to different intensities of anger responses (Humprecht et al., 2020).

But before measuring the effect of framing on emotion in short videos, a more fundamental question must be answered: What constitutes framing in short-video communication?

From Text to Image to Video

Since the Gutenberg era, scholarly traditions have tended to assign cultural primacy to the written word, treating it as the finest vehicle for human thought. This orientation naturally

influenced early framing research, which focused primarily on textual information and treated narrative rhetoric and organizational patterns in news headlines or political propaganda as the main entry points for identifying frames. As a result, framing research developed a relatively mature methodological tradition centered on textual modality. With the “visual turn” in the contemporary humanities and social sciences, however, the role of images, videos, and other visual materials in meaning production has received increasing attention (Mitchell, 1994). At the same time, the growing prevalence of dynamic visual information in contemporary media content, especially on short-video platforms, makes it increasingly inadequate to identify persuasive frames solely through linguistic representation.

In fact, the associative logic of visual communication differs substantially from the linear reasoning logic of textual communication. Visuals rely more heavily on parallel and heuristic information processing, whereas text is typically processed in a more sequential manner (Messaris, 2003). Moreover, within the same unit of time, images can convey more information than text, allowing viewers to access meaning in a relatively automatic and holistic way (Coleman, 2010). Visuals therefore possess an independent and relatively complete meaning system of their own (Kress & van Leeuwen, 2020). Compared with text, they are also more capable of attracting attention and eliciting stronger emotional responses (Iyer & Oldmeadow, 2006; Mendelson, 2004).

Against this background, visual frames have gradually been conceptualized as analytical objects that can be compared and coded. Rodriguez and Dimitrova (2011), for example, distinguish visual framing across four levels—denotative, stylistic-semiotic, connotative, and ideological—thereby moving the study of visual framing from intuitive description toward a more systematic analytical approach.

Although visual frames have been confirmed to be distinct from textual frames, current framing research still focuses on static visual materials, such as the relationship between photojournalism and text (Rodriguez & Dimitrova, 2011). Apart from a small number of qualitative interpretations, existing research lacks a systematic approach to identifying and

measuring continuous, temporal, and multimodal visual frames in dynamic videos (Fazeli et al., 2023; Geise & Xu, 2025). Existing studies have begun to process visual features in videos with the help of computational methods, such as identifying scientific conspiracy theory videos through symbolic features such as color and luminance (K. Chen et al., 2022), or examining how nonverbal elements displayed by politicians in debate videos affect voters' immediate impressions (Bucy & Joo, 2021). However, most of these studies have focused on local visual features and have yet to consider video as a communication unit with a continuous temporal sequence and an overall organization of meaning. They therefore lack a more structural understanding of its internal visual framing (Fazeli et al., 2023; Geise & Xu, 2025). Based on this, this paper attempts to construct an automated visual-frame identification method for large-scale short videos by using Chinese TikTok videos related to the sealing of drug-use records in China as sample materials and examines the main visual frames present in this issue accordingly.

Consistency and Conflict in Multimodal Framework Combinations

As a multimodal media form consisting of both visual and textual frames, short videos provide new analytical material for understanding how framing affects the expression of anger in the comment section, as well as presenting more complex theoretical issues. Unlike static multimodal media, visuals and texts in short videos do not function independently of each other, but are in continuous interaction. Based on modal characteristics, visuals and texts assume different but complementary functions in the communication process (Kress & van Leeuwen, 2020). Visual materials usually have a significant advantage in attracting attention and evoking emotions, and their holistic nature and high information density enable them to set the context and emotional tone in a very short time. In contrast, texts are better at providing causal chains and normative evaluations, transforming contextual information into attributable and judgmental meanings through linear logical deduction.

Powell et al. experimentally compared the effects of image frames and text frames both individually and in combination. They found that, when the two are presented separately, images tend to influence audiences' attitudes and behavioral intentions more than text. When graphics

and text are presented together, the audience's response is not simply the sum of the two modal effects; instead, consistency or conflict between the cognitive frames of images and text changes the overall framing effect (Powell et al., 2015). Their research exposes a multimodal interaction problem: the orientations of visuals and text within a communication unit are not always the same. Frames and narrative expressions are also not always coherently arranged in short videos. Thus, in highly multimodal communication environments such as short videos, emotional effects cannot be inferred solely from textual stance or frame content, but should be understood in the context of how visuals and texts combine, reinforce, or counteract each other.

Research in recent years has further advanced this question in more practical settings. For example, consistent graphics and text in climate change news contribute to news salience, while audiences are often more likely to prioritize the perspective implied by a picture when graphics and text are inconsistent (Mosallaei & Feldman, 2024). Inconsistency in graphic sentiment in charity awareness posts, especially the combination of negative-sentiment images and positive-sentiment text, significantly reduces audience engagement (Kwon et al., 2022), while consistency in graphic themes in restaurant review posts promotes messaging and audience engagement (Ceylan et al., 2024). Traditional news media also take into account the interplay between visual and textual frames in their print reporting in order to use visual symbols to corroborate or even amplify the core position presupposed by the text (Ehmer & Kothari, 2018). Overall, existing research has demonstrated two points more clearly. First, the visual frame has independent validity and cannot be relegated to an appendage of the textual frame. Second, visual and textual frames may either be consistent and mutually reinforcing or be misaligned and competing within the same communication unit, which has a significant impact on communication effects.

However, while existing empirical studies have shown that visual materials are often more dominant in attracting attention and evoking emotions, and multimodal framing studies have shown that combinations of different modalities can affect attributions of responsibility and policy judgments, more direct empirical tests are still lacking. Specifically, it remains unclear

how visual and textual frames form comparable relational structures within a continuous and dynamic communication object such as a short video and whether these structures are correlated with expressions of anger in the comment section. Based on this, this paper considers the combined relationship between visual and textual frames as the focus of analysis and poses the following research questions:

RQ1: What are the distributional characteristics of dominant visual frames and dominant textual frames in short videos related to the issue of “sealing drug abuse records”?

RQ2: What are the combination patterns of dominant visual frames and dominant textual frames in a single short video?

RQ3: Is there a significant correlation between different combinations of visual-textual frames and the intensity and distribution of angry expressions in the comment section? Are their framing effects superimposable?

Method

Description of Data Sources and Sample Construction

In this paper, the Douyin platform (China’s version of TikTok) is selected as the data source for the case discussion. As one of the largest short-video platforms in China in terms of user size, Douyin plays an important role in the dissemination of public issues and the diffusion of emotions. The cases examined in this study are directly related to the revision of the *Law of the People’s Republic of China on Penalties for Administration of Public Security*. In June 2025, the third review draft of the revision proposed that “records of violations of public security administration shall be sealed,” and the officially revised law took effect on January 1, 2026. Between November and December 2025, the issue heated up rapidly in public discourse as media reports focused on the fact that “drug records are also subject to sealing.” Against the backdrop of China’s long-standing crackdown on the drug trade, the proposal sparked widespread public concern over public safety and public controversy, which in turn inspired heated debate on the topic. Therefore, the short-video content and the interactions in its comment sections on Douyin provide a suitable empirical field for examining the relationship between the multimodal framing

of short videos and the expression of anger in comment sections.

In this paper, relevant issue videos on the Chinese Douyin platform and their first-level comment texts are taken as the research object, and a self-constructed Python crawler is used for data collection. Considering that public issues usually go through a cycle of public attention from rapid heating up to the decline of attention, this paper combines the event's dissemination rhythm on the platform and sets the data crawling period from November 26, 2025, to February 20, 2026 (Downs, 2016). After the initial data capture was completed, the videos were screened based on keyword relevance, content integrity, and sample availability. Samples irrelevant to the research topic, duplicate uploads, incomplete content, or videos for which comment information could not be effectively accessed were excluded. Ultimately, 449 relevant short videos and 419,126 comments were obtained.

Computational Framework Analysis

In this paper, we encode emotion in short videos from two dimensions: narrative text and visual modality, and form the corresponding dominant frame variables at the video level. It should be noted that this paper clearly distinguishes between “textual modality” and “visual modality” in the operationalization, and the textual content in the picture is not counted as visual modality, but extracted separately and incorporated into the textual material package (Fazeli et al., 2023). The purpose of this process is to avoid the text in the picture from directly interfering with the visual coding, so that the visual frame can rely more on non-textual visual resources, such as characters, scenes, objects, actions, compositions and atmospheres, to be recognized.

At the level of narrative frames, this paper first transcribes the voice, subtitles and extra description in short videos to form video text. Subsequently, the transcribed text is cleaned and standardized, and encoded on the basis of retaining the original semantic structure. Combining the open coding results and sample features, this paper summarizes the higher-order frames at the text level into three categories: intensifying frames, informational, and mitigating frames. Among them, the intensifying frames refer to the text's organization of the issue as a problem that requires vigilance, condemnation, or quick action through the expressions of risk, prevention,

injustice, connivance, accountability, cost, social harm, and institutional failure, thus strengthening the sense of tension and emotional arousal. Texts that explicitly highlight potential risks, emphasize the responsibility of actors or institutional arrangements, use explicitly cautionary, accusatory, or condemnatory language, and push audiences to form negative moral judgments are classified as intensifying frames. Informational framework means that the text mainly assumes the functions of introducing events, supplementing background, explaining the system, explaining rules, clarifying boundaries and stating facts, and does not aim at promoting the escalation of emotions or actively cooling down the temperature. Texts that mainly provide facts, explain policies, explain procedures, clarify misinterpretations or supplement general information, and that do not obviously reinforce risk alerts, attribution of responsibility and moral condemnation, and that do not obviously depress the tension of the issue through procedural and standardized expressions, are all categorized as informational frameworks. The mitigating framework refers to the text's emphasis on institutional boundaries, procedural control, rule constraints, professional response and misinterpretation clarification, etc., to organize the issue as a problem that can be understood and dealt with in accordance with the formal procedures, so as to reduce the sense of risk and emotional tension. Where the text clearly highlights procedural and normative expressions such as "not leaving the matter unattended", "there is still a system of control" and "it should be understood in accordance with legal procedures", and is mainly oriented toward cooling down, de-risking, or stabilizing cognition, the text is categorized as a mitigating frame. Due to the small sample size, coding was conducted by two communication studies students, and the coding consistency was good (Cohen's kappa = 0.85), indicating that the categorization results have good reliability.

At the level of visual framing, instead of representing an entire short video with a single static screenshot, this paper treats the video as a dynamic material composed of successive visual segments. This paper draws on the process of frame extraction, de-emphasis and semantic analysis adopted by VisTopics, and bases visual frame recognition on continuous frames rather than isolated images. The method compresses 11,070 frames into 6,928 de-emphasized images in

452 NBC News videos and further identifies 35 visual themes, showing its scalability in large-scale visual material analysis (Lokmanoglu & Walter, 2025). Combined with the characteristics of the object of this study, the visual frame recognition will be divided into the following steps: first, frame extraction is performed for each video at a fixed frequency of 1 frame per 1 second in order to preserve as much as possible the continuous visual content of the video and its temporal distribution. Second, a perceptual hash value is computed for the extracted neighboring frames to identify highly similar frames that occur consecutively, and these similar frames are grouped into the same visual segment. Third, a representative frame is retained in each group of similar frames as the semantic recognition object for that visual segment. Fourth, the representative frame is input into the LLM or image description model to generate structured visual description text, such as character type, scene type, action relationship, emotional expression, symbols, camera distance and overall atmosphere. Fifth, based on this, deductive coding is performed on each representative frame according to the pre-established visual frame class list, and the coding results are mapped back to the original video timeline. The full code of the program is backed up on GitHub. (Examples of coding are shown in Table 1.)

In contrast to the text coding, the higher-order visual frames are also summarized into three categories: intensifying, informational, and mitigating frames. Among them, intensifying frames mainly represent warning, deterrence, anomalization, painful consequences, and a high-pressure atmosphere, such as police lights, cordons, arrest associations, masking and coding, shadowed characters, drug paraphernalia, tragic consequences, and scenes of heavy victimization. Informational frames are mainly manifested in general information images, ordinary oral broadcasts, news broadcasts, routine interviews, and informational shots, i.e., they are mainly responsible for information transmission and background introduction, but do not obviously strengthen the sense of risk, nor do they take the initiative to take on the function of cooling down. The mitigating frame is mainly expressed in the law page, institutional doorplate, blue-background response page, procedural information, and professional information screens in normative spaces, which convey the meaning of controllable treatment of the issue through

institutionalized, proceduralized, and orderly visual resources. Finally, this paper calculates the duration ratio of intensifying, informational, and mitigating visual frames in a single video and identifies the dominant visual frame as the one with the longest duration ratio.

Multimodal Relational Variables

After the textual and visual coding is completed, this paper further constructs the relationship variables between the two at the video layer. Firstly, we compare whether the dominant textual frame is consistent with the dominant visual frame, and accordingly form the combination types of consistent, inconsistent and partially consistent. At the same time, this paper also retains the specific combination variables of textual and visual frames to identify different types of multimodal collocation, and the subsequent statistical analysis will use these relationship variables to examine the correlation between multimodal structure and anger expression in the comment section.

Anger Measurement Approach

In this paper, “expression of anger” is defined as the commenter’s apparent dissatisfaction, condemnation, rebuke, or resentment toward an actor or institutional arrangement, and the moral reflection of a perceived norm violation (Leach et al., 2026). Comments that only express other negative emotions such as helplessness, sadness, disappointment, fear, or mockery, or that contain only negative evaluations without forming a clear direction of anger are not included in the category of anger expression. This definition draws on the ideas of established research on the operationalization of anger expression in social media texts (Monge & Laurent, 2024).

In the specific coding process, this paper firstly extracts 10,000 comment samples from all the comment data to construct a manually labeled set, and then determines the comments one by one according to the coding manual. The coding manual clearly stipulates that any angry words, such as “furious”, ‘angry’, “outrageous”; expressions with strong reprimanding meaning, such as “How dare you?” “That’s disgusting”, and text that reinforces the angry tone with auxiliary symbols such as consecutive question marks, exclamation marks, and emoticons were judged to

be angry expressions. The manual labeling was done independently by two coders, who were trained and pretested before the formal labeling. The Cohen's Kappa coefficient of the two coders on 10,000 randomly selected pre-labeled samples was tested to be 0.87, which was extremely consistent. For entries where there was disagreement, agreement was negotiated through discussion.

In order to identify the intensity of anger expression in the full sample data, this study uses BERT, a linguistic pre-trained model based on the bidirectional Transformer structure, for binary classification of comment section text, and fine-tunes the pre-trained BERT model based on the content analysis results as a means to create the optimal model.

The training parameters of the BERT model based on the labeled set are shown in Table 2, where `Validation_split` represents the proportion of the validation set in the overall dataset, `Batch_size` reflects the overall characteristics of the sample data, `Epochs` represents the number of iteration rounds of the model training, and `Learning_rate` is the learning rate that determines the convergence speed of the model. The AdamW optimizer is used for optimization. According to the results of the emotion classification task, the precision rate and recall rate of the model tend to be close to the ideal state, and the emotion classification model can be considered more accurate (see Table 2).

Results

Sample and Variable Description

At the level of sample size, a total of 449 relevant videos were included in this study, corresponding to 419,126 first-level comments, with an average of 933.47 effective comments per video, which shows that the topic has high interaction intensity and visibility in the platform. After identification by the machine learning model, 48,973 angry comments are visible, accounting for about 11.68% of all first-level comments. The video-level anger rate has a mean of 0.099 and a median of 0.087.

Overall about 12% of the first-level comments show clear expressions of anger, but there are significant fluctuations in the level of anger between different videos. In terms of dependent

variable setting, this paper takes a single short video as the unit of analysis, aggregates the results of anger identification at the comment level into an anger rate indicator at the video level, and takes it as the core dependent variable. In terms of informational variable setting, this paper constructs two core variables: dominant visual frame and dominant textual frame. The visual frame is determined based on the time share of different visual contents in the video, which is categorized into intensifying, informational and mitigating, with sample sizes of 68 (15.1%), 189 (42.1%) and 192 (42.8%), respectively. Textual frames were coded based on oral, subtitle, and extra-informational descriptions, and were similarly categorized as intensifying, informational, and mitigating, with sample sizes of 78 (17.4%), 111 (24.7%), and 260 (57.9%) items, respectively. Overall, both the visual and textual dimensions were dominated by informational and mitigating frames, with a relatively low percentage of intensifying frames. The results are shown in Figure 1.

Differences in Anger Expression Across Combined Modes

Although the contingency table in Figure 2 shows that visual and textual frames often appear in tandem, with more consistency and less conflict, it remains unknown whether different frame combinations correspond to different levels of anger. To examine whether different combinations of visual and textual frames correspond to different levels of anger expression in the comment section, this paper uses the Kruskal–Wallis test to compare between-group differences across the nine combinations. The results show that video-level anger rates differ significantly across combinations, $\chi^2(8) = 50.98, p < .001$.

Descriptive statistics further show a clear hierarchical differentiation in anger rates. When both visual and textual frames are oriented toward mitigation, the anger rate is the lowest ($M = 0.064$), and the median anger rate is only 0.035, substantially lower than that of the other combinations. Although the anger rate is relatively high when both visual and textual frames are oriented toward intensification ($M = 0.142$), it is still slightly lower than that of the combination of an informational visual frame and an intensifying textual frame.

As shown in the figure below, textual intensification appears to be more directly

associated with elevated anger expression. Regardless of whether the visual frame is informational or intensifying, anger rates remain high once the textual frame enters the intensifying category. This suggests that textual emphasis on risk and responsibility may drive anger expression more directly than visual stimulation alone. In contrast, the dual de-escalation combination shows the clearest cooling effect, with comment-section anger rates significantly lower when both visual and textual mitigating frames are adopted. This trend is even more robust when examined through the median.

In summary, there is a significant association between different combinations of visual and textual frames and anger expression in the comment section, with a clear directional pattern: combinations containing intensifying visual and textual frames are more likely to correspond to higher levels of anger, whereas combinations in which both frames favor mitigation are associated with significantly lower anger rates. The results are shown in Figure 3.

Two-Factor ANOVA

Having confirmed that different visual–textual combinations correspond to significantly different anger rates, this paper further examines whether this difference arises from the particular combinations themselves or from the accumulation of the separate effects of visual and textual frames. To this end, a two-way ANOVA was conducted with video-level anger rate as the dependent variable and dominant visual frame, dominant textual frame, and their interaction as predictors.

The results show that the main effect of visual framing is significant, $F(2, 440) = 17.74, p < .001$, and the main effect of textual framing is also significant, $F(2, 440) = 5.51, p = .004$, whereas the interaction between the two is not significant, $F(4, 440) = 0.28, p = .893$. As shown in Table 3, anger rates differ significantly across visual frames and across textual frames, but visual and textual framing do not form a stable interaction-enhancing mechanism. In other words, although the nine combinations display clear high–low differentiation in descriptive terms, this pattern does not imply that certain combinations produce special effects beyond the sum of the two modal effects.

To verify this interpretation, this paper further estimated a main-effects model without the interaction term. The results show that the main effect of visual framing remains significant, $F(2, 444) = 13.89, p < .001$, as does the main effect of textual framing, $F(2, 444) = 5.54, p = .004$. This suggests that the effects of visual and textual frames on anger expression do not depend on the interaction term, but instead operate stably as relatively independent factors. Combined with the descriptive statistics above, the intensifying or mitigating tendency in visuals and texts changes the anger rate independently, and the variation across combinations is mainly the result of the superposition of the two effects at the video level. In interpretive terms, visual and textual modalities have independent effects on anger expression in the comment section, but these effects do not exhibit significant interactive amplification and are instead closer to linear superposition.

Robustness Test

To test the robustness of the above findings, two sets of additional analyses were conducted. First, because the video-level anger rate is a proportional variable bounded between 0 and 1, this paper uses a fractional logit model in addition to the baseline OLS model to re-estimate the relationship between visual frames, textual frames, their interaction terms, and the anger rate. Second, the paper incorporates platform-context variables such as likes, followers, and posting time into the baseline linear model to examine whether the effects of visual and textual frames remain after controlling for account attributes and dissemination timing.

The fractional logit model yields a ranking of predicted anger rates across visual–textual combinations that is broadly consistent with the baseline OLS model. Among them, the informational visual–intensifying textual combination has the highest predicted anger rate, 0.145, 95% CI [0.111, 0.186]; the intensifying visual–intensifying textual combination ranks second, 0.142, 95% CI [0.110, 0.180]; and the mitigating visual–mitigating textual combination is the lowest, 0.064, 95% CI [0.053, 0.077]. The predicted values for the remaining combinations range from 0.099 to 0.128. This result suggests that the main conclusion of this paper—that combinations containing intensifying text are generally associated with higher anger rates, whereas dual de-escalation combinations are associated with the lowest anger rates—does not

substantially change when a model more appropriate for proportional outcomes is used.

Discussion

This study takes Douyin short videos related to the issue of the sealing of drug use records in China as its case, and, by combining computational framing analysis with anger detection in comment sections, examines the ways in which visual frames and textual frames are combined in short videos, as well as the relationship between this multimodal structure and anger expression in the comment section. Based on the preceding data analysis results, this study arrives at three findings. First, the relationship between visual frames and textual frames is not a random juxtaposition, but rather shows a significant same-direction association, indicating that multimodal meaning production in short videos may be manifested more as coordinated organization rather than mutual disconnection. Second, different combinations of visual and textual frames do in fact correspond to different levels of anger rates in the comment section, and such combinations exert a certain influence on the elicitation of anger in the comment section. Third, the effects of visual frames and textual frames on anger expression are mainly reflected in the parallel accumulation of their respective main effects, rather than in significant statistical interaction effects.

At the methodological level, the method illustrated above may help advance multimodal framing research in the context of dynamic short videos. Although using only the dominant frame with the longest duration as the analytical material cannot fully capture the visual characteristics of a video in emotional terms, and although a certain key frame in a video may be more capable of provoking anger than all other frames combined, this method may still be helpful for the study of visual framing, especially for exploring intermodal relationships in large-scale data, and may prove useful as artificial intelligence tools gradually become more reliable (Fazeli et al., 2023; Lokmanoglu & Walter, 2025; Peng et al., 2024).

At the theoretical level, by answering RQ1 and RQ2, this study finds that, within the same event, the emotional tendencies of visual frames and textual frames can indeed exist independently while at the same time influencing audiences. They are usually organized into a

whole oriented in the same direction (Geise & Xu, 2025; Powell et al., 2015). This means that the multimodal structure in short videos is not a rhetorical device attached to content from the outside; it is necessarily also part of meaning production.

This study also extends the discussion of framing effects from cognitive judgment to the level of overt emotional expression. The study by Kühne and Schemer shows that different news frames can induce different discrete emotions, and that intensifying frames increase the accessibility of information related to punishment and strengthen preferences for punitive measures (Kühne & Schemer, 2015). The results of this study echo this line of thinking and extend it into the public, interactive, and observable emotional arena of platform comment sections. Powell et al.'s experimental study finds that when images and text are presented separately, images produce stronger framing effects than text, whereas when images and text appear together, congruence and conflict alter the overall effect (Powell et al., 2015). The results of this study are broadly consistent with this judgment. However, judging from the present findings, a more cautious interpretation is not that “text is necessarily more important than visuals,” but rather that intensifying narratives in text come closer to explicitly articulating moral condemnation and imaginings of punishment, and therefore are more likely to be directly transformed into overt anger in the comment section. Visual modality, by contrast, is closer to the prior setting of situation, sense of threat, and emotional tone. Precisely for this reason, the two are statistically manifested as a parallel accumulation of main effects rather than a stable mechanism of interaction enhancement. These results further show that, in short videos, emotion is not triggered instantaneously by a single modality, but is gradually organized through the joint operation of visual and textual attribution of responsibility.

This finding also helps us better understand anger expression on platforms. The reason why intensifying textual frames are more likely to correspond to higher anger rates is precisely that they usually use narrative resources such as risk, precaution, breakdown, indulgence, sacrifice, and accountability to organize the issue into one that demands rapid moral judgment. By contrast, mitigating frames lower audiences' perceptions of loss of control, indulgence, and

injustice through expressions emphasizing institutional boundaries, procedural controllability, and the continued operation of rules, thereby reducing the space for the release of anger expression. In other words, anger in short-video comment sections is not merely the natural spillover of the controversy of the issue itself; it is also the result of how content organizes responsibility attribution and controllability.

Limitations

Finally, several limitations of this study still need to be noted. Anger expression in the comment section is only visible overt emotion, rather than the entirety of audiences' internal emotional states, and this study cannot directly observe how silent viewers feel about video content. Second, this study focuses on a single issue and a specific time window. Therefore, the current conclusions are better understood as an empirical finding concerning highly controversial governance issues, rather than as a general law applicable to all public issues. At the same time, due to the constraints of China's censorship system, the data collection of this study was subject to considerable limitations (King et al., 2013). Finally, what this study reveals is a systematic association between multimodal framing and anger expression, rather than strict causal identification in the strong sense. Future research may further expand to different issues, different platforms, and different cultural contexts, and combine experimental designs, longitudinal tracking, or more fine-grained interaction data to continue examining how visuals and text jointly shape emotional expression in platform environments.

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Note1. In this paper, DeepL was used in the translation process, while this paper used GPT for generating the R language used for data analysis. Extra, in the content this paper describes a method for large-scale video visual frame analysis using LLM. All interpretations, analysis, and final decisions remain valid. GPT was used to polish the language of this article before the final version was submitted.

Table 1. Visual coding examples

Visual frames	Coding indicators	Examples
Intensifying frame	Images emphasizing threat, punishment, suffering, or moral urgency, such as police lights, handcuffs, obscured faces, dark or tense scenes, drug-related objects, and other warning-oriented visual elements.	
Informational frame	Images mainly providing factual background or issue explanation, such as ordinary talking-head shots, news broadcasts, routine interviews, and general descriptive footage without obvious emotional intensification or de-escalation.	
Mitigating frame	Images emphasizing rules, order, and institutional control, such as legal documents, official response pages, institutional signs, procedural charts, and professional informational settings that suggest the issue remains governable and bounded.	

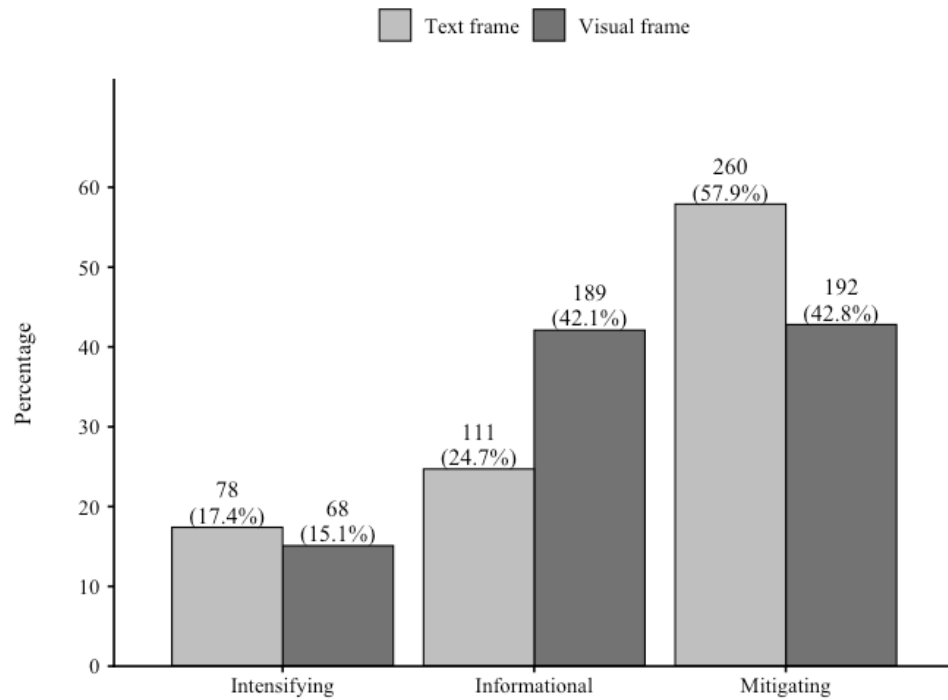
Table 2. Training parameters and classification performance of the BERT sentiment classification model

Validation_split	0.1
Batch_size	8
Epochs	5
Learning rate	2e-5
Optimizer	AdamW
Precision	0.976
Recall	0.968
F1	0.972
Accuracy	0.993

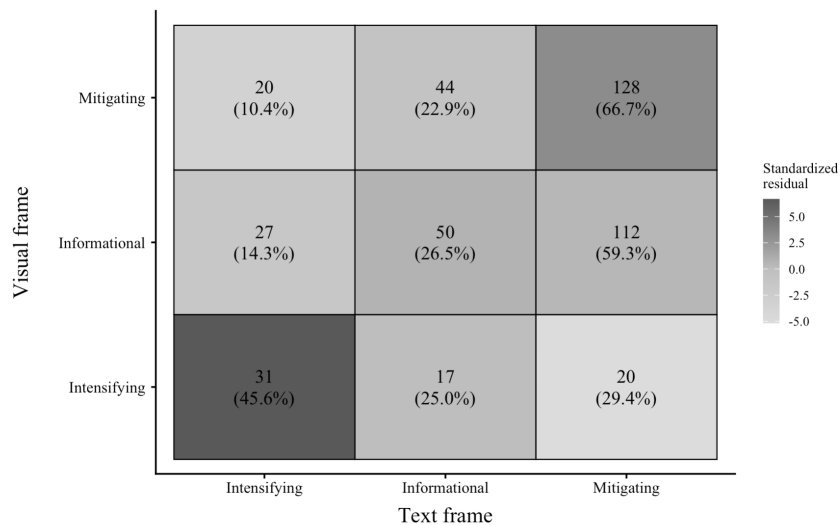
Table 3. Two-way ANOVA results for video-level anger rate

Effect	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>	<i>p</i>	Partial η^2
Visual Frame	0.252	2	0.126	17.74	< .001	.075
Text Frame	0.078	2	0.039	5.51	.004	.024
Visual $\times$ Text	0.008	4	0.002	0.28	.893	.003
Residual	3.124	440	0.007			
Total	3.462	448				

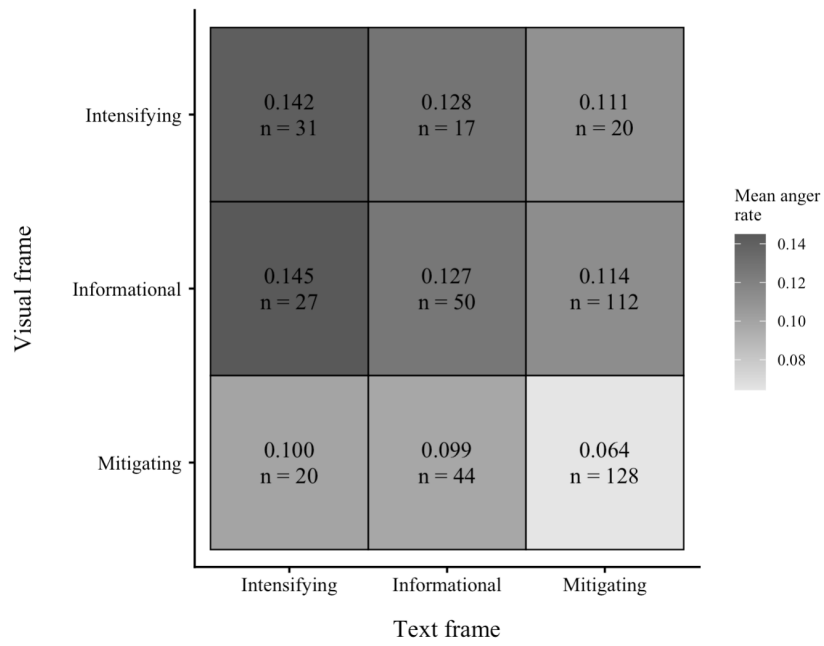
Note. $N = 449$. The table reports results from the full two-way ANOVA model including the main effects of visual frame and text frame, as well as their interaction. Partial η^2 indicates effect size.

Figure 1. Distribution of dominant text and visual frames

Note. Bars represent the percentage distribution of dominant text and visual frames across 449 videos. Numbers above bars indicate frequency and percentage. Light gray bars denote text frames, and dark gray bars denote visual frames.

Figure 2. Distribution of visual–text frame combinations

Note. Cells report frequency and row percentage. Color shading represents standardized residuals from the chi-square test, where darker cells indicate combinations occurring more often than expected and lighter cells indicate combinations occurring less often than expected. $\chi^2 = 50.171$, $df = 4$, $p < .001$, Cramér's $V = .236$.

Figure 3. Mean anger rate across visual–text frame combinations

Note. Cells report the mean anger rate and sample size for each visual–text frame combination. Darker shading indicates a higher mean anger rate. Kruskal–Wallis test: $\chi^2 = 50.984$, $df = 8$, $p < .001$.